

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Although Vision–Language Models (VLMs) constitute a promising substrate for zero-shot goal-conditioned value functions, their reliance on frozen pre-trained representations constrains both distributional generalization and temporally grounded inference. We introduce VITA, a zero-shot value-function learning method that improves these two capabilities through meta-learned test-time adaptation. During inference, a lightweight adaptation module undergoes a gradient update on a meta-learned self-supervised objective, so that each test-time update is optimized to refine subsequent value estimation. Because these updates are applied sequentially along a trajectory, VITA stores trajectory history in its parameters, thereby mitigating the temporal-reasoning deficiencies of static representation-based estimators. To reduce shortcut learning, we further introduce a dissimilarity-based sampling strategy that preferentially selects semantically heterogeneous trajectory segments during training. On real-world robotic manipulation tasks, VITA transfers from a single training environment to diverse out-of-distribution tasks, environments, and embodiments, surpassing the state-of-the-art zero-shot approach based on autoregressive VLMs. Finally, we show that VITA’s zero-shot value estimates can serve as reward-shaping signals for offline reinforcement learning, producing multi-task Meta-World policies whose performance exceeds that of policies trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs), trained on large-scale and weakly structured web corpora without direct human supervision, have exhibited unusually broad generalization across heterogeneous tasks and domains (Radford et al., 2021; Alayrac et al., 2022). By contrast, even though recent work has substantially advanced generalist policy learning for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), prevailing methods remain comparatively constrained because they typically depend on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) have consequently emerged as a promising mechanism for enabling agents to learn in simulated environments, with applications that span robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet, even if such models can synthesize realistic visual trajectories in diverse and open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), a central unresolved question persists: *how can agents effectively learn from videos at scale without relying on human supervision?*

One research trajectory seeks to learn policies directly from expert visual trajectories by inferring latent actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). Although these inferred actions can, in principle, be translated into executable controls at deployment time, the corresponding grounding problem remains practically difficult. A complementary line of work instead aims to learn a universal *goal-conditioned value function*, which can be used as a zero-shot reward-shaping signal for reinforcement learning (RL) or as a supervision signal for imitation learning (Chen et al., 2021; Ma et al., 2022). Under this formulation, goal-conditioned value estimation amounts to inferring, from visual observations together with a natural-language task description, the extent to which an agent has progressed toward task completion (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work has used pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping by exploiting the similarity between task descriptions and visual observations within a shared multimodal representation space, thereby grounding progress estimates in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Such approaches, however, lack the temporal context required to distinguish visually similar states that correspond to different stages of a task, as in the difference between folding and unfolding a shirt. Autoregressive VLMs (Alayrac et al., 2022) address temporal conditioning by placing an entire visual trajectory in the prompt, but this strategy inherits a bias toward monotonically increasing predictions from the chronologically ordered data used during pre-training (Ma et al., 2024). Because both architectural families depend on pre-trained representations for zero-shot prediction (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), their generalization and temporal reasoning remain circumscribed (Pătrăucean et al., 2023). Recent efforts have therefore pursued large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demonstration-based reward specification (Sontakke et al., 2023) as alternative routes toward more generalizable value-function learning.

In this work, we introduce VITA, a zero-shot value-function learning method that augments generalization and temporal reasoning through test-time adaptation, and that outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. Rather than depending on additional large-scale pre-training or task-specific demonstrations at deployment, VITA adapts online to the semantic and temporal structure of each trajectory. During inference, a lightweight adaptation module is updated, with negligible overhead, by a gradient step on a meta-learned self-supervised loss (Sun et al., 2024); the meta-objective is constructed so that each such update improves value-function estimation (Finn et al., 2017). Sequential adaptation over a test trajectory causes the estimator to encode trajectory history directly in its parameters (Sun et al., 2024), thereby addressing the temporal-reasoning limitations of prior work. To further discourage shortcut learning (Geirhos et al., 2020), we introduce a dissimilarity-based sampling strategy that, as supported empirically, encourages reliance on semantic rather than merely chronological cues. Our evaluation examines three central capabilities of value-function estimation: generalization under distribution shift, discrimination between expert and non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation method that increases the distributional generality and temporal inferential capacity of contrastive VLMs for zero-shot value-function estimation, without requiring task-specific demonstrations or additional large-scale pre-training.
- VITA transfers from a single training environment to heterogeneous out-of-distribution robotic manipulation tasks, environments, and embodiments, while outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- When used for reward shaping, VITA’s zero-shot value estimates produce offline RL policies on the Meta-World multi-task learning benchmark (MT10) that surpass policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task-progress estimation can be cast as goal-conditioned value-function estimation, with reward interpreted as the degree of task completion. We therefore formulate the learning problem as the prediction of completion progress from vision–language representations. Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, mapping an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar estimate of progress toward goal satisfaction. We assign $V(o_t; g) = 0$ to the initial task state and $V(o_t; g) = 1$ to completion. In expert demonstrations, progress is commonly aligned with temporal position under the assumption that such trajectories move monotonically toward the goal (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is specified by normalized timestep indices, $V^{\pi_E}(o_t; g) = \frac{t}{T}$, where T denotes trajectory length; these normalized indices consequently provide supervision for learning a goal-conditioned value function V .

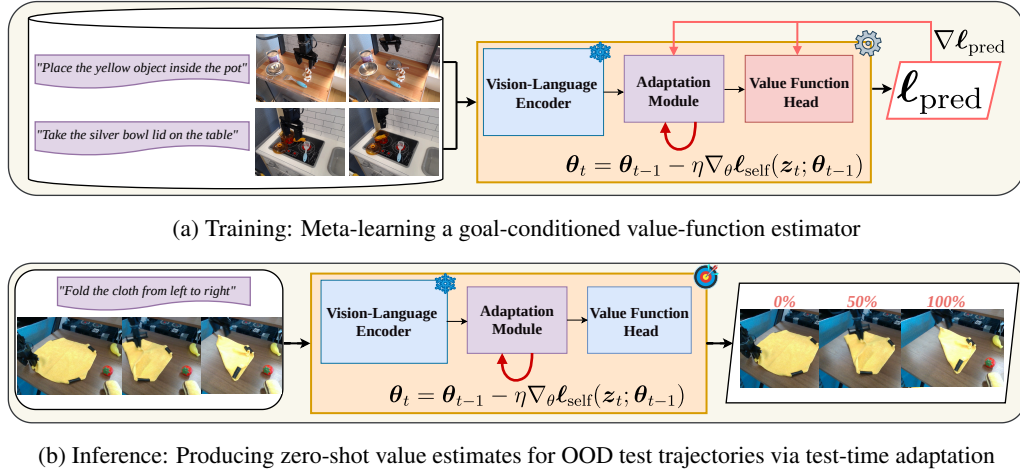


Figure 1: **Overview of VITA.** VITA learns a goal-conditioned value function through meta-learning and obtains zero-shot generalization to out-of-distribution trajectories through test-time adaptation.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a test-time adaptation paradigm that recasts inference as a self-supervised learning problem (Goodfellow et al., 2016), updating model parameters on each unlabeled test instance. Although initially developed for static image classification (Sun et al., 2020), subsequent work extends TTT to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} receives a gradient update at each timestep under a self-supervised objective. Unlike conventional sequence architectures, which store temporal history in activation states such as recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT deposits history into the parameters of f_{adapt} through sequential inference-time updates. The parameters of f_{adapt} thereby function as an implicit memory that preserves temporal order while accumulating trajectory context. Sun et al. (2024) further propose meta-learning the self-supervised task within a gradient-based meta-learning framework (Finn et al., 2017); in this setting (Sun et al., 2024), learnable linear projections parameterize the self-supervised loss, which is optimized not for its own reconstruction accuracy but for the downstream prediction loss obtained after a test-time update. We adopt this formulation for goal-conditioned value-function learning, so that temporal context is acquired through sequential updates and the self-supervised task is meta-learned specifically to improve value estimation rather than being fixed a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value-function estimator is organized around three components: (1) a multimodal encoder that extracts joint vision-language representations from visual trajectories and their task descriptions; (2) a test-time adaptation module updated by a self-supervised loss whose parameters are meta-learned to improve value estimation; and (3) a regression head that maps adapted representations to scalar value predictions.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We employ a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), to obtain representations of both visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate the corresponding representations at each timestep t , yielding a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and where ϕ_v and ϕ_g denote the visual and language encoders, respectively. Because CLIP is pre-trained with a contrastive objective that aligns paired image-text inputs within

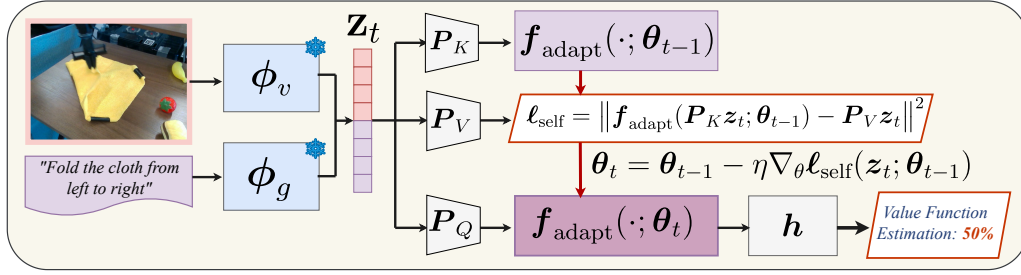


Figure 2: **Test-time Adaptation.** During inference, at each timestep t , the adaptation module f_{adapt} is updated by a gradient step on the meta-learned self-supervised loss ℓ_{self} , thereby encoding temporal history in its parameters.

a shared representation space (Radford et al., 2021), visual observations that are semantically nearer to goal completion tend to lie closer to the representation of the corresponding goal description.

3.1.2 TEST-TIME ADAPTATION

To make multimodal representations responsive to both semantic content and temporal context, we introduce an adaptation module f_{adapt} following the test-time training paradigm described in 2.2. At inference, the parameters of f_{adapt} are updated online through a gradient step on a meta-learned self-supervised loss ℓ_{self} . This objective is specified as a reconstruction loss involving learnable linear projections P_K and P_V , whose parameters are meta-learned during training so that the resulting test-time adaptation improves value estimation. Specifically, with $P_K \in \mathbb{R}^{d' \times 2d}$ producing a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ producing the reconstruction target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} immediately before the test-time update. At each timestep t , these parameters are updated according to

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Because the updates are applied sequentially, f_{adapt} implicitly retains information from previous visual observations and thereby encodes temporal context. Section 4.6.2 provides empirical evidence that incremental parameter updates of the value-function estimator are more effective than trajectory-level updates.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into the adaptation space $\mathbb{R}^{d'}$. The projected representation is then processed by the adaptation module f_{adapt} and subsequently by a regression head h , which produces the predicted goal-conditioned value:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is instantiated as a two-layer multilayer perceptron (MLP), consistent with value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). As described in Section 2.1, the network is trained with normalized progress labels y_t from expert demonstrations, using a supervised prediction loss ℓ_{pred} given by the mean squared error between the predicted values $V(z_t; g)$ and targets y_t . During inference, the regression head h remains fixed, whereas the adaptation module f_{adapt} is updated online under the self-supervised objective, enabling zero-shot generalization without task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value-function estimator through gradient-based meta-learning, optimizing the self-supervised loss ℓ_{self} so that the resulting test-time adaptation decreases the supervised prediction

loss ℓ_{pred} (Sun et al., 2024). Training uses diverse sub-trajectories selected by dissimilarity-based sampling, a procedure intended to attenuate shortcut learning while increasing reliance on semantic and temporal evidence.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep by the self-supervised loss ℓ_{self} . Following the gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} , thereby optimizing the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . Formally, the total training objective combines the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , weighted by a scalar λ . The learned parameters θ_0 subsequently initialize inference-time adaptation, allowing the value-function estimator to learn how its internal representations should change at test time in order to improve value estimation.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories frequently contain consecutive frames with substantial redundancy, a phenomenon also emphasized in prior work on video action recognition (Wang et al., 2018). This redundancy can induce the value-function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), particularly by overfitting to recurrent late-stage visual patterns. To mitigate this failure mode, we introduce a dissimilarity-based sampling strategy that constructs mini-batches from the most visually diverse sub-trajectories within each trajectory. The procedure increases intra-batch variance and functions as a form of importance sampling that privileges semantic diversity.

Formally, for a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we first extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , which yields a candidate set $\mathcal{W}$. A diverse subset of size k is then characterized by the maximization of pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i denotes a sub-trajectory of length w_{tr} . Directly solving Eq. 4 is intractable because it requires enumerating all $\binom{|\mathcal{W}|}{k}$ subsets. We therefore approximate the objective with an efficient scoring heuristic that assigns each window $w \in \mathcal{W}$ a score equal to its aggregate dissimilarity from all other windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Consequently, computing a diversity score s for every window and selecting the k highest-scoring candidates yields a heuristically diverse subset in polynomial time. As shown by the complexity analysis in Appendix G, this additional computation is negligible relative to model training. Section 4.6.1 empirically compares dissimilarity-based sampling with full-trajectory sampling and shows that the former improves discrimination between expert and non-expert visual trajectories.

4 EXPERIMENTS

Our evaluation assesses VITA along three core capabilities expected of goal-conditioned value functions: (i) value-estimation generalization for real-world robotic manipulation under task, environment, and embodiment shifts; (ii) discrimination between expert and non-expert visual trajectories in real-world manipulation; and (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated benchmark for multi-task robot learning.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions from BridgeData V2 (Walke et al., 2023), a dataset spanning diverse manipulation tasks, environments, and robot embodiments. Specifically, we use a curated subset of 2,986 expert demonstrations for pick-and-place manipulation; this subset deliberately excludes folding, sweeping, and stacking tasks.

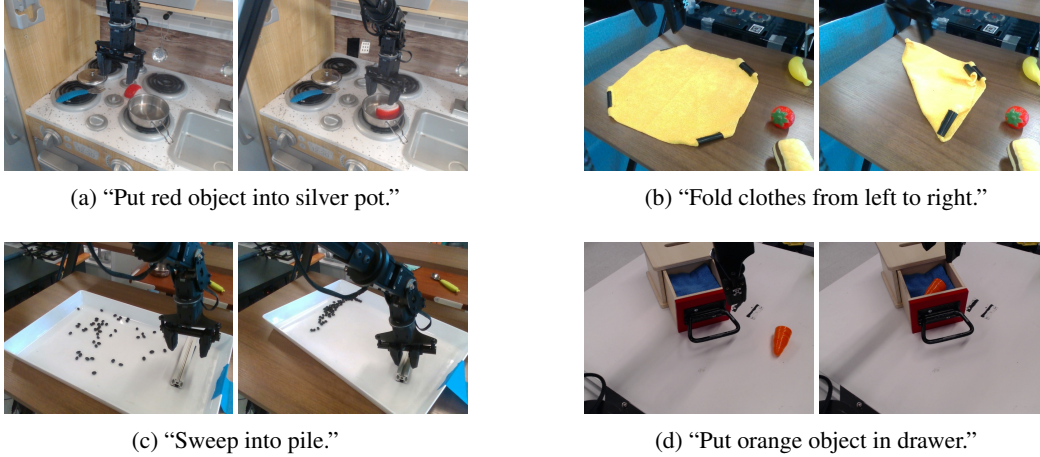


Figure 3: Representative visual trajectories paired with task descriptions across distinct distributional regimes. (a) In-distribution. (b, c) Environment shift. (d) Joint embodiment and environment shift.

All training demonstrations are collected with a single robot embodiment (WidowX 250) across 4 ToyKitchen environment configurations. We reserve an additional 287 expert demonstrations as the in-distribution test set, denoted `tk_pnp`; Appendix A provides representative training examples. OpenCLIP ViT-B/32 serves as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$. During training, dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, we apply one gradient step ($t_{\text{ep}} = 1$) using λ_{self} with learning rate $\eta = 0.1$. Additional implementation details and hyperparameter sweeps appear in Appendix D.

4.2 BASELINES

We compare VITA with value-function estimators built on contrastive and autoregressive VLMs. VLM-CL derives zero-shot value estimates from the cosine similarity between CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, whereas CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same frozen representations. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value-function method based on autoregressive VLMs. In our experiments, GVL-0S denotes the zero-shot setting in which the VLM receives only the test trajectory and task description, while GVL-1S denotes a one-shot in-context setting in which a full shuffled trajectory from the training distribution, together with its progress labels, is supplied as an exemplar. Following GVL, we use Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) with the proposed prompt template. We also evaluated Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to emit value estimates. Appendix D contains additional implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

We evaluate whether progress estimators generalize across novel environments, tasks, and robot embodiments by using BridgeData V2 subsets curated to vary along precisely these axes. Environment shifts alter scene layout or background: for instance, `lm_pnp` is a pick-and-place task situated in front of a laundry machine, whereas `td_fold`, `ft_fold`, and `rd_fold` involve cloth folding on distinct surfaces. `ms_sweep` further introduces a long-horizon sweeping task in a confined tray. Embodiment shifts are evaluated with the DeepThought robot, whose morphology and camera perspective differ from the training embodiment (WidowX 250). `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) retain the in-distribution environment while changing embodiment, whereas `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) introduce both embodi-

Table 1: VOC scores for value-function estimation under distributional shift. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Simultaneous Environment and Embodiment Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

ment and environment shifts. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 because of its size limitation. Appendix B gives the full evaluation-dataset specification. We assess value-function estimators with *Value Order Correlation* (VOC) (Ma et al., 2024), which measures the alignment between predicted task-completion progress and the chronological ordering of frames within a visual trajectory. Formally, let $p_1, \dots, p_T$ denote predicted progress values for a trajectory of T frames, and let $r_t = t$ denote the temporal index. VOC is the Spearman rank correlation ρ_s between predicted values and temporal indices.

The results in Table 1 indicate that VITA and CLIP-GRU consistently exceed GVL-0S and GVL-1S, underscoring both the limitations of pre-trained autoregressive VLMs for this setting and the importance of preserving temporal order during progress estimation. Although both temporally informed methods perform well, VITA outperforms CLIP-GRU on most evaluation datasets (6 of 10). Moreover, both VITA and CLIP-GRU substantially outperform CLIP-based baselines without temporal modeling (CLIP-FT, VLM-CL, VLM-RM), demonstrating the practical significance of encoding temporal history. CLIP-FT is competitive with GVL on the in-distribution task, yet its performance deteriorates under out-of-distribution conditions, while VLM-CL and VLM-RM remain weak, plausibly because they lack temporal structure. For GVL-0S and GVL-1S, in-context examples sometimes improve performance, sometimes provide no measurable benefit, and in some cases degrade estimation quality. Although both GVL variants perform well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`), they do not attain comparable performance on stacking (`dt_tk_stack`, `dt_ft_stack`) or pick-and-place (`dt_tk_pnp`, `lm_pnp`) tasks, suggesting that the autoregressive VLM used in GVL may be biased toward folding-like robotic manipulations. By contrast, VITA maintains consistent performance across task types and distribution shifts, indicating stronger generalization than in-context learning.

On the long-horizon `ms_sweep` task, all methods except CLIP-FT experience reduced performance, yet VITA still obtains the highest VOC score. Its advantage over CLIP-GRU in this setting suggests that encoding temporal history through test-time adaptation is more effective for long-horizon robotic manipulation than representing that history solely through recurrent hidden states. Under embodiment shift, VITA also generalizes strongly. In `dt_tk_pnp`, where the robot embodiment changes while the environment and task family remain aligned with training, both VITA and CLIP-GRU exceed their in-distribution performance, indicating that learned temporal context can transfer across embodiments. Conversely, GVL-1S performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, suggesting that few-shot in-context learning does not reliably generalize under embodiment shifts in our evaluation setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we examine VITA’s robustness by testing whether it distinguishes expert visual trajectories from suboptimal ones, which should receive lower progress scores. We therefore compare model predictions on expert and scripted (non-expert) trajectories collected from the

same in-distribution BridgeData V2 setting. The scripted trajectories are generated in the ToyKitchen environment with a heavily randomized controller (Walke et al., 2023), under the same embodiment and environmental configuration as the expert demonstrations. They comprise pick-and-place tasks whose object categories may overlap with the training set, including general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution with respect to task, embodiment, and environment, their behavioral structure departs from the smooth and monotonic progress characteristic of expert demonstrations. Appendix C presents examples from the scripted dataset. Because the scripted datasets are produced by a randomized controller, they are not intended to encode calibrated levels of suboptimality; nonetheless, an effective value-function estimator should assign lower VOC scores to suboptimal trajectories than to expert trajectories evaluated in the same setting. We quantify this discrimination using *BinVOC*, a binary metric defined as $1 [\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when expert trajectories obtain higher VOC than suboptimal trajectories and 0 otherwise.

We report discriminative performance as BINVOC averaged over all scripted datasets. Expert-trajectory performance (VOC_{exp}) is measured on the in-distribution test set `tk_pnp`, whose pick-and-place tasks are comparable to those in the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination, consistently assigning higher VOC scores to expert trajectories across all tasks. VITA’s advantage over CLIP-GRU suggests that recurrent hidden-state memory may be more susceptible to temporal-shortcut overfitting than implicit memory acquired through sequential test-time updates. The weaker performance of CLIP-based baselines (VLM-CL, VLM-RM) is plausibly attributable to their absence of temporal modeling. CLIP-FT is more reliable, but its failure on one task indicates sensitivity to object-distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We evaluate the value-function estimator on Meta-World MT10 (Yu et al., 2020), which contains ten diverse robotic manipulation tasks, including pick-and-place, door opening, and pushing. This experiment tests whether VITA, trained on real-world robotic data, can provide zero-shot reward shaping for offline RL in a simulated benchmark. For each Meta-World MT10 task, we generate 20 expert visual demonstrations using Meta-World’s expert policies, and we use the goal-conditioned value estimator to define a dense reward for each visual observation. Policies are trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL setting. We repeat expert-data generation and policy training across 10 random seeds. After training, each multi-task policy is evaluated on 20 rollouts per task, and the average return is computed across episodes for every (task, seed) pair. Following the evaluation protocol of Agarwal et al. (2021), we report the interquartile mean (IQM) across all pairs together with 95% stratified bootstrap confidence intervals. Appendix E provides the training details and evaluation protocol.

Table 2 reports MT10 performance for the baselines under the preceding evaluation protocol. VITA attains the highest IQM return (0.815), exceeding all CLIP-based baselines. A scaled direct comparison with GVL is infeasible because GVL relies on Gemini-1.5, a proprietary autoregressive VLM whose inference cost is prohibitive in RL settings, while the open-source alternatives examined in preliminary tests did not produce reliable progress estimates. We additionally report the fuzzy-logic dense reward provided by Meta-World (META-WL), which achieves 0.779. Although META-WL is strong, VITA surpasses it, indicating that a value estimator trained on real-world data can transfer effectively to simulated reward shaping.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We empirically validate dissimilarity-based sampling by comparing how alternative sub-trajectory sampling strategies affect VITA’s ability to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, which in our setting encourages overfitting to global temporal shortcuts. Random sampling increases diversity by selecting sub-trajectories uniformly, but does not explicitly enforce semantic diversity. Dissimilarity-based sampling, by contrast, selects sub-trajectories that maximize pairwise dissimilarity and therefore directly promotes semantically diverse adaptation sig-

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-OS	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discriminative success in separating expert demonstrations from scripted robot demonstrations, measured by average BinVOC across 5 in-distribution scripted datasets. (b) Offline RL performance on the Meta-World MT10 benchmark, reported as interquartile mean (IQM) with 95% stratified bootstrap CIs.

nals. This proposed strategy outperforms both full-trajectory and random sampling in discriminating expert from non-expert visual trajectories; Appendix F.1 provides further details.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how alternative test-time adaptation strategies for incorporating temporal context affect value-function estimation. VITA performs stepwise adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} and thereby encoding trajectory history directly in the estimator parameters; this mechanism preserves both temporal order and accumulated context. We compare VITA with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update computed over the complete trajectory and thus forms a batched update that averages loss across frames, discarding temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step, so adaptation to the current visual observation cannot carry temporal information forward; and (iii) *explicit memory (TTT-EX)*, which likewise resets the adaptation module at every step but updates it using a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently outperforms the trajectory-level, memoryless, and explicit-memory variants, demonstrating the effectiveness of incremental temporal-context accumulation for value-function estimation. Appendix F.2 contains the full analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been used as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), typically through frame-level similarity scores that do not explicitly model temporal structure. Other work fine-tunes CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to produce dense rewards, but the required supervision limits zero-shot applicability. RoboCLIP (Sontakke et al., 2023) supports one-shot reward specification without fine-tuning, yet its rewards are sparse and demonstrations remain necessary. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on scale for generalization, while ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding and a small number of in-domain demonstrations. In contrast, VITA adapts a pre-trained contrastive vision–language encoder at inference time, capturing both semantic and temporal context through test-time adaptation without requiring large-scale multimodal pre-training or additional domain-specific demonstrations for generalization. Because the method is tied to the adaptation mechanism rather than to a particular encoder, VITA could be applied to any pre-trained vision–language representation that encodes a universal value function, improving its generalization and temporal reasoning through meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022) exploit in-context learning over trajectories and have

been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) mitigates monotonic bias from ordered pre-training data by shuffling frames at inference, but this intervention removes temporal order, which is essential for temporal reasoning and for distinguishing visually similar yet temporally distinct states. Our approach instead preserves chronological order while capturing temporal context through test-time adaptation, thereby avoiding this shortcut.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers a parameter-efficient route for adapting VLMs without full fine-tuning. Prompt-based methods optimize model parameters at test time according to task context, although the mechanisms underlying their adaptation remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations may support test-time adaptation, motivated by evidence that CLIP encodes human-interpretable concepts revealed through mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also served as a reward signal for test-time parameter optimization via reinforcement learning (Zhao et al., 2024). A related alternative is test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep under a self-supervised objective. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training permits direct online adaptation during inference without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our results show that test-time adaptation enables robotic-manipulation value functions to generalize across shifts in task, environment, and embodiment. Beyond this generalization, VITA distinguishes expert from non-expert visual trajectories and supplies zero-shot reward-shaping signals, thereby supporting downstream RL and imitation-learning applications. By updating sequentially along a trajectory, VITA encodes history in its parameters and captures temporal and semantic context more effectively than both CLIP-based baselines and autoregressive-VLM in-context learning. Its advantage over CLIP-GRU on expert–non-expert discrimination and offline RL suggests that implicit memory produced by sequential test-time updates generalizes more effectively, for zero-shot value-function estimation, than explicit recurrent hidden-state encoding of temporal history. Although pre-trained autoregressive VLMs (GVL) possess broad generalization capabilities, they are not explicitly trained for progress estimation, a task that requires temporal reasoning over visual trajectories. In our implementation, the inference-time overhead of per-timestep adaptation remains negligible because only a lightweight adaptation module is updated at test time, leaving real-time applicability unaffected.

6.2 LIMITATIONS AND FUTURE WORK

Although test-time adaptation improves zero-shot progress-estimation generalization across unseen tasks and environments in our experiments, it may remain challenging in settings characterized by high execution variability or substantially longer horizons. Moreover, even though the lightweight adaptation module makes the computational overhead negligible, updating a value-function estimator at every timestep could introduce deployment risks in some real-time scenarios. Future work should therefore investigate alternative test-time adaptation mechanisms for vision–language models, particularly when agents are trained inside world models. Extending VITA to real-time closed-loop control and more complex RL environments is likewise an important direction. Because VITA is agnostic to the underlying vision–language encoder, it could improve the generalization and temporal reasoning of domain-specific pre-trained multimodal representations through meta-learned test-time adaptation. Finally, while our experiments indicate that dissimilarity-based sampling mitigates shortcut learning, a theoretical account of how diversity-based sampling alters shortcut reliance remains open. Such an analysis is beyond the scope of the present contribution, which introduces a value-function estimator that meta-learns test-time adaptation to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

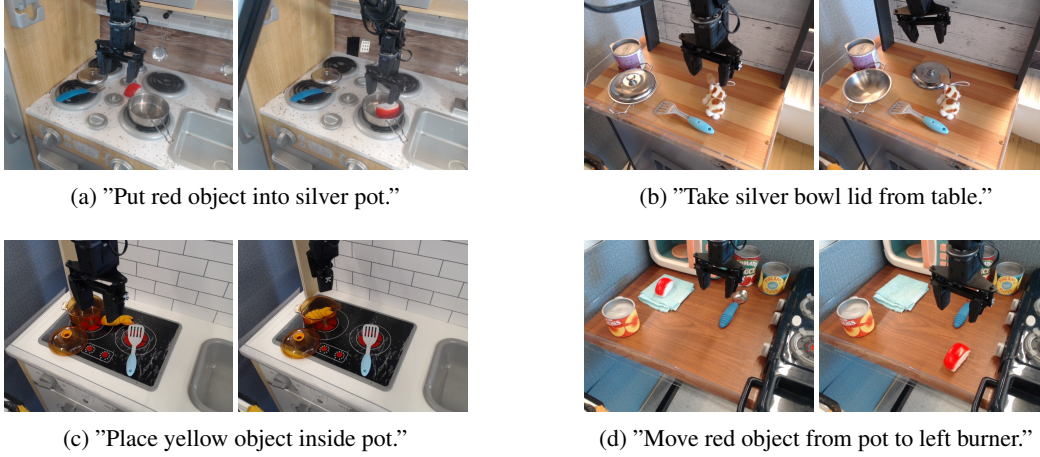


Figure 4: Each subfigure presents the initial and terminal frames from an expert training demonstration, together with its natural-language task description. The demonstrations are collected across four distinct ToyKitchen environments.

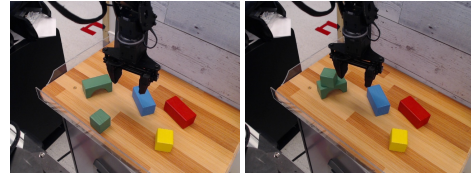
B EVALUATION DATASETS

Table 3: Evaluation-dataset descriptions specifying task type, environment, and robot embodiment.

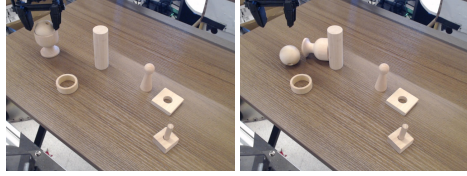
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

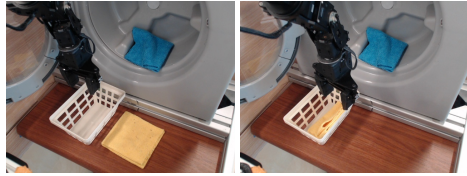


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure displays the initial and terminal frames from an evaluation trajectory under embodiment shift, accompanied by its natural-language task description. The top row retains the ToyKitchen environment while replacing the robot with DeepThought, whereas the bottom row additionally introduces new environments.



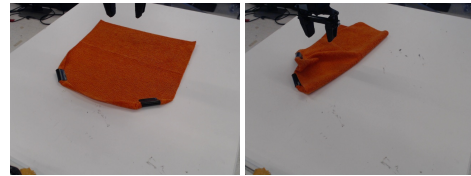
(a) lm_pnp: "Place blue cloth inside washer."



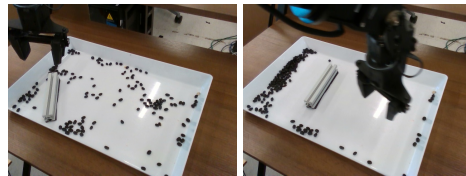
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure displays the initial and terminal frames from an environment-shift evaluation trajectory, together with its natural-language task description. All tasks use the same robot embodiment while varying the visual and structural properties of the environment.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure presents the initial and terminal frames from a scripted evaluation trajectory collected in the ToyKitchen environment with the WidowX 250 robot. All tasks involve pick-and-place behavior over different object categories; because public annotations are unavailable, we use generic task descriptions.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone. Each visual observation and task description is encoded with the CLIP vision and text encoders, respectively, to obtain paired representations. Based on validation performance, we use a joint concatenation-based representation rather than an element-wise product. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. The model is trained with AdamW using learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine learning-rate schedule with 10% warmup. We use batch size 32 and pad all trajectories to a maximum length of 120 frames, matching the longest sequence in the training set. The self-supervised loss weight λ_{self} in the total training objective is set to 0.5 according to validation performance. We train for 5 epochs because validation VOC generally plateaus early and extended training yields no additional improvement. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and number of selected windows $k = 8$. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time, only the temporal adaptation module f_{adapt} is adapted, using the same projection dimension ($d' = 64$) as during training. We perform a single gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. This configuration is selected from a sweep over learning rates

Table 4: Effect of test-time learning rate η and adaptation-step count t_{ep} on VITA validation performance as measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

$\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation-step counts $\{1, 5, 10\}$, using the validation performance reported in Table 4. The sweep shows that both (i) one update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and (ii) multiple updates with a smaller learning rate ($\eta = 0.01$, $t_{\text{ep}} = 10$) attain comparable validation performance under test-time training. We choose the former because it requires fewer adaptation steps while producing slightly better results.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate η of 0.1, as selected by the hyperparameter tuning described above. CLIP-FT follows VITA’s architecture but removes the adaptation module and self-supervised loss: it uses a frozen CLIP encoder followed by a linear projection and a two-layer MLP for task-progress prediction, and it is trained with standard supervised regression rather than meta-learning or test-time adaptation. To increase expressivity, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt describing the environment. CLIP-GRU preserves VITA’s frozen CLIP encoder and regression head h , but replaces the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) that explicitly encodes temporal history through recurrent hidden states. The GRU contains one layer with hidden size 64, and its final hidden state is passed to the regression head h . For both GVL-0S and GVL-1S, we use the latest Gemini 1.5 Pro release, gemini-1.5-pro-latest (Gemini Team et al., 2024), whereas the original GVL implementation used the earlier gemini-1.5-pro. We also evaluated GVL with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations used the prompt template introduced in the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with a configuration held fixed across all tasks. Expectile regression uses $\tau = 0.7$, advantage weighting temperature is 3.0, and the clipping threshold is 100.0. The actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies are trained for 100,000 gradient steps and evaluated every 10,000 steps on 20 rollouts per task with horizon 150. The implementation follows d3rlpy, with configuration: expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. IQM averages the middle 50% of scores, thereby reducing sensitivity to outliers relative to the mean or median. To quantify statistical uncertainty, we estimate 95% confidence intervals using a stratified bootstrap with 10,000 resamples, stratified by task. The same procedure is applied to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC results across scripted datasets for alternative training-time sampling strategies used in test-time adaptation. A checkmark denotes successful discrimination between expert and suboptimal trajectories. Sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We further validate dissimilarity-based sampling by comparing alternative sub-trajectory sampling strategies and their influence on the model’s discrimination between expert and suboptimal trajectories. In Table 5, Disc@VOC is computed for 5 scripted datasets, indicating whether a model assigns higher average VOC to expert trajectories (tk_pnp) than to each scripted evaluation dataset. We compare three strategies: (1) full-trajectory sampling, which follows prior work (Sun et al., 2024) by applying adaptation over all overlapping sub-trajectories of length w_{tr} with stride s ; (2) random sampling, which uniformly selects k sub-trajectories of length w_{tr} from each video; and (3) dissimilarity-based sampling, which constructs a candidate set of sub-trajectories of length w_{tr} using stride s and selects k sub-trajectories that maximize pairwise dissimilarity in representation space.

The results show that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discriminative performance. Selecting fewer windows ($k = 4$) of size $w_{tr} = 8$ yields the same outcome, whereas shortening the sub-trajectory length to $w_{tr} = 4$ reduces effectiveness. The full-trajectory baseline performs worst, confirming that adaptation over all consecutive windows can overfit to global temporal cues. Random sampling improves over full-trajectory sampling but remains less reliable than dissimilarity-based selection. These findings emphasize that both sub-trajectory diversity and semantic locality are important for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC scores for progress estimation under distributional shift. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Joint Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We evaluate four adaptation strategies: (1) TTT-TR, which performs a single offline gradient update computed over the full trajectory; because this batched update averages the loss across all frames, it discards temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates only on the current visual observation, thereby producing a memoryless variant that prevents temporal information from being carried forward; (3) TTT-EX, which also resets the adaptation module at every step but updates using a local window of recent frames whose size matches the meta-training window ($w_{\text{tr}} = 8$); because gradients are averaged across this window, temporal ordering within the window is discarded; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep and therefore forms implicit memory through the sequential accumulation of temporal history in adapter parameters. We performed the same test-time hyperparameter sweeps for all variants, as described in Appendix D. VITA, TTT-RS, and TTT-TR achieve optimal performance with learning rate 0.1 and one adaptation step, whereas TTT-EX performs best with learning rate 1.0 and one adaptation step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX achieve similar performance, implying that batched updates over either a short explicit window (TTT-EX) or the full trajectory (TTT-TR) behave comparably to the memoryless baseline (TTT-RS). By contrast, VITA consistently outperforms all three variants, emphasizing that sequential per-timestep updates are crucial for effective value-function estimation in real-world robotic manipulation. Under implicit memory, updating over a local frame window would incorporate the most recent observations twice, once explicitly through the window and once implicitly through the accumulated parameters. Conversely, less frequent non-sequential updates would lose information by omitting intermediate frames. The implicit-memory test-time training paradigm therefore requires sequential per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set with size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without incurring the exponential cost of enumerating all $\binom{N_c}{b}$ subsets, as in Eq. 4, we use a scoring heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all other windows, i.e., the row sum of the distance matrix. The number of candidate windows is $N_c = T - w_{\text{tr}} + 1$, and each window is represented by a flattened feature vector of dimension $m = w_{\text{tr}} \cdot d$, where d denotes the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, requiring $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$ operations.

In our experiments, training trajectories are relatively short, with mean length 69 and maximum length 120. With batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to a single forward and backward pass through our model, while still promoting the selection of diverse windows.